Inputs

Electrode connector

Four differential filters

Common reference + bias

File: Inputs.kicad_sch

Frontend

ADS1299-4 acquisition

250 samples/s/channel

Reference / supply bypass

File: Frontend.kicad_sch

Radio

ESP32-C3 wireless data

BLE streaming

UART for bench programming

File: Radio.kicad_sch

Power

Removable protected 2S pack

3.3 V buck / 5 V analog LDO

External charging only

File: Power.kicad_sch

4 channels + common reference + bias electrode; 250 samples/s/channel nominal.

Protected removable 2S battery, 6.4–8.4V; external charging only.

BLE data via ESP32-C3; UART header is for bench programming only.

BENCH PROTOTYPE: validate using a simulator before any human connection.

Disconnect electrodes before charging, wired programming, or grounded instruments.

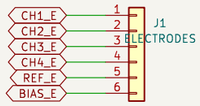
Sheet: /
File: EEG.kicad_sch

**Title: Four-channel wireless EEG learning prototype**

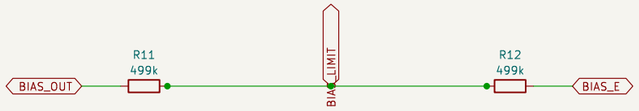
| Size: A3 | Date: | Rev: A – bench prototype |
|---|---|---|
| KiCad E.D.A. 9.0.9.1 | | Id: 1/5 |

# ELECTRODES / INPUT PROTECTION

J1: channels 1–4, reference, bias




Bias electrode: two independent series resistors



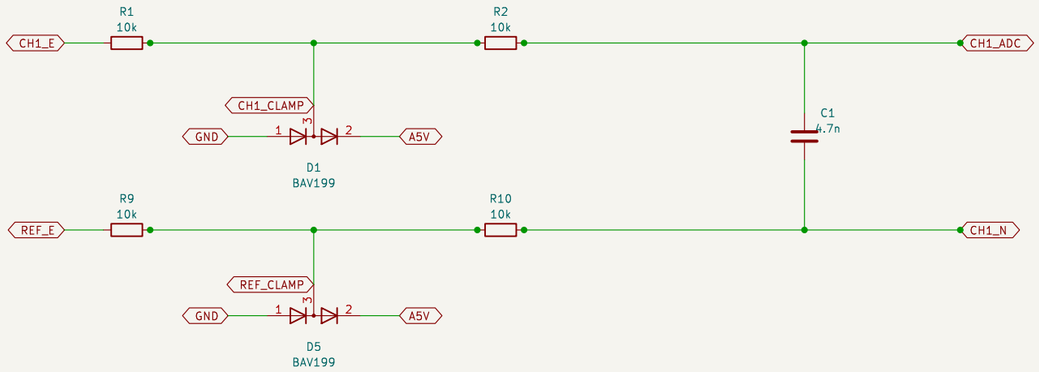

[I0] REF compares all channels; BIAS is a driven output.

J1 has no circuit-ground electrode. Trace CH1_E to CH1_ADC.

BIAS current path: R11 + R12 = 998 kOhm; bench validation pending.

## CHANNEL 1 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below




[I1] R1/R9 limit current toward the clamps; R2/R10 add isolation.

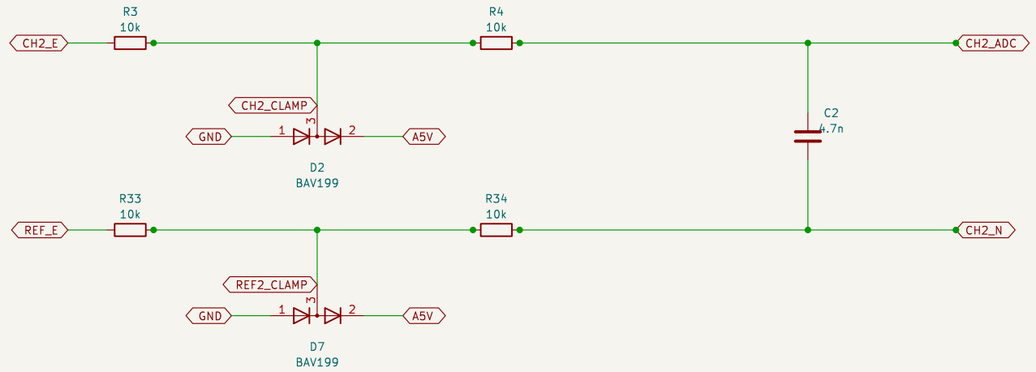
D1/D5: low-leakage rail clamps; leakage creates input offset.

C1 bridges P/N: 40 kOhm differential + 4.7 nF gives about 847 Hz.

This is high-frequency attenuation, not the final EEG bandpass.

## CHANNEL 2 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below




[I2] REF_E fans out into four separate protected branches.

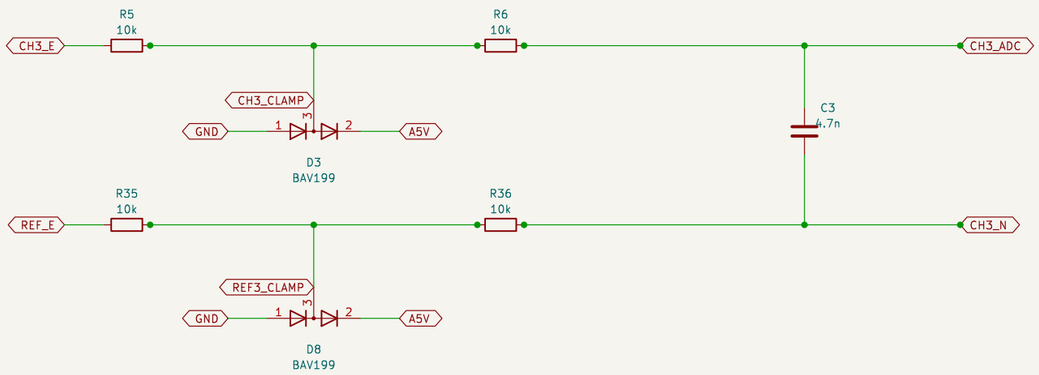
Each channel has its own negative input and differential capacitor.

A disturbance on the shared reference can affect several channels.

Channels 3/4 repeat this topology. Clamps do not certify safety.

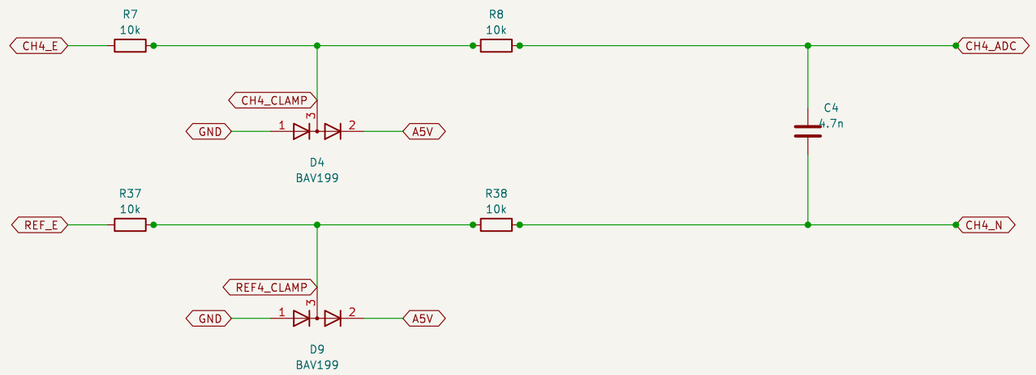
## CHANNEL 3 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below




## CHANNEL 4 / DIFFERENTIAL FILTER

Positive electrode above; shared REF branch below




20 kOhm per input branch / 4.7 nF differential / rail clamps are not a certified patient interface

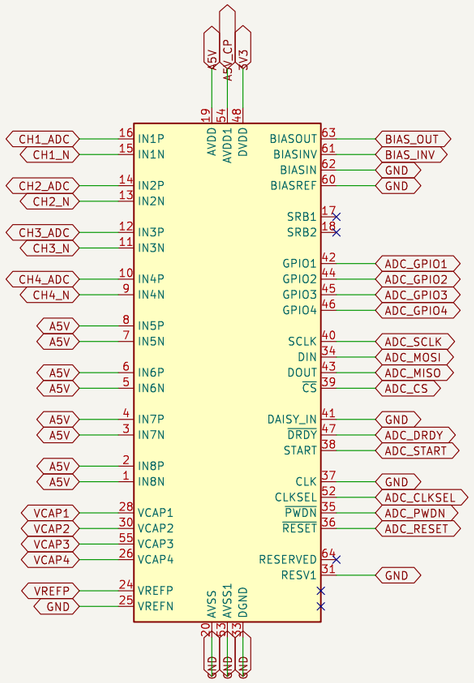
| | |
|---|---|
| Sheet: /Inputs/ | |
| File: Inputs.kicad_sch | |
| **Title: EEG learning prototype – Inputs** | |
| Size: A3 | Date: 2026-10-12 |
| KiCad E.D.A. 9.0.9.1 | **Rev: A – bench prototype** |
| | Id: 2/5 |

# ADS1299-4 / ACQUISITION CORE

Four differential inputs from Inputs sheet; SRB1/SRB2 not used

[F1] U1 converts four differential channels simultaneously.
24-bit format does not imply 24-bit measurement accuracy.

U1
ADS1299-4

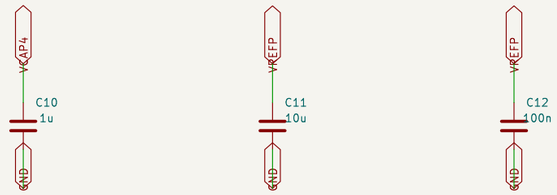



## REFERENCE / VCAP BYPASS

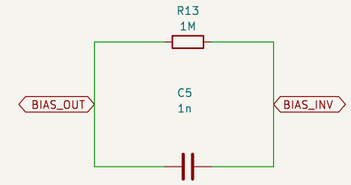
Follow TI placement guidance; C6 polarity matters




[F2] VCAP capacitors support internal circuits; not spare supplies.
C11/C12 steady the ADC conversion reference, not the REF electrode.
Use specified values and short local return paths; C6 is polarized.

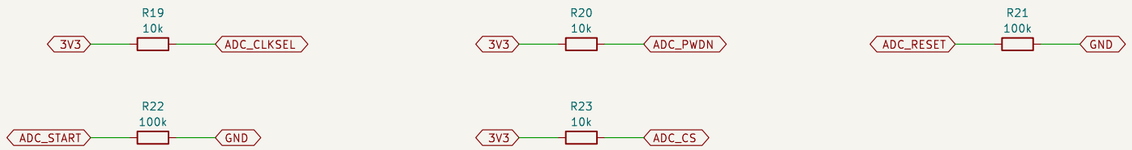



Bias amplifier feedback




[F3] R13 || C5 shapes bias feedback.
RC scale ~159 Hz; actual loop stability needs testing.
Use internal bias reference; enable after bench checks.

Hardware defaults

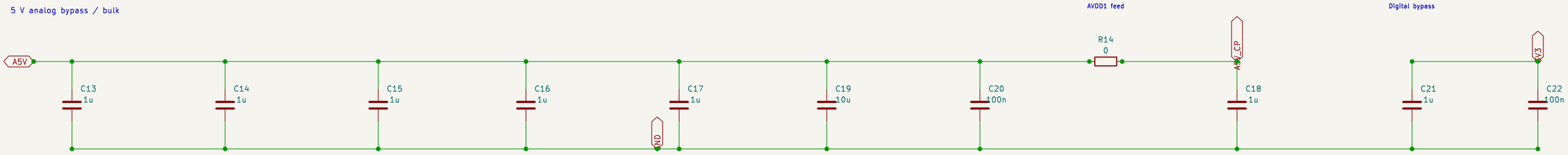



[F4] Pulls define startup: RESET/START low, CS high.
CLKSEL/PWDN high; firmware must configure the ADC.

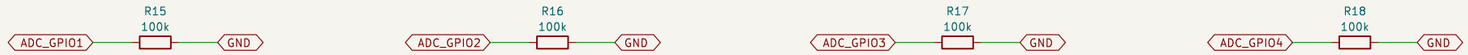
5 V analog bypass / bulk

AVDD1 feed

Digital bypass



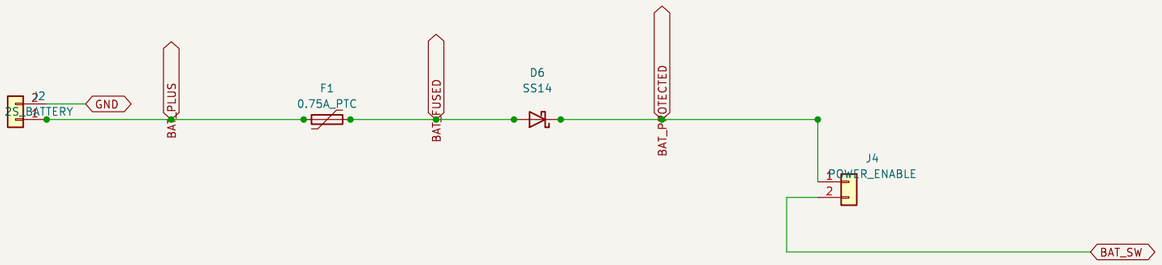

Unused GPIO pull-downs




[F5] Local bypass supplies transient current. R14 is a removable link, not a filter.

| | | |
|---|---|---|
| Sheet: /Frontend/ | | |
| File: Frontend.kicad_sch | | |
| **Title: EEG learning prototype — Frontend** | | |
| Size: A3 | Date: 2026-10-12 | **Rev: A — bench prototype** |
| KiCad E.D.A. 9.0.9.1 | | Id: 3/5 |

## BATTERY INPUT / DISCONNECT




[P1] F1: thermal overcurrent response, not a precise cutoff.

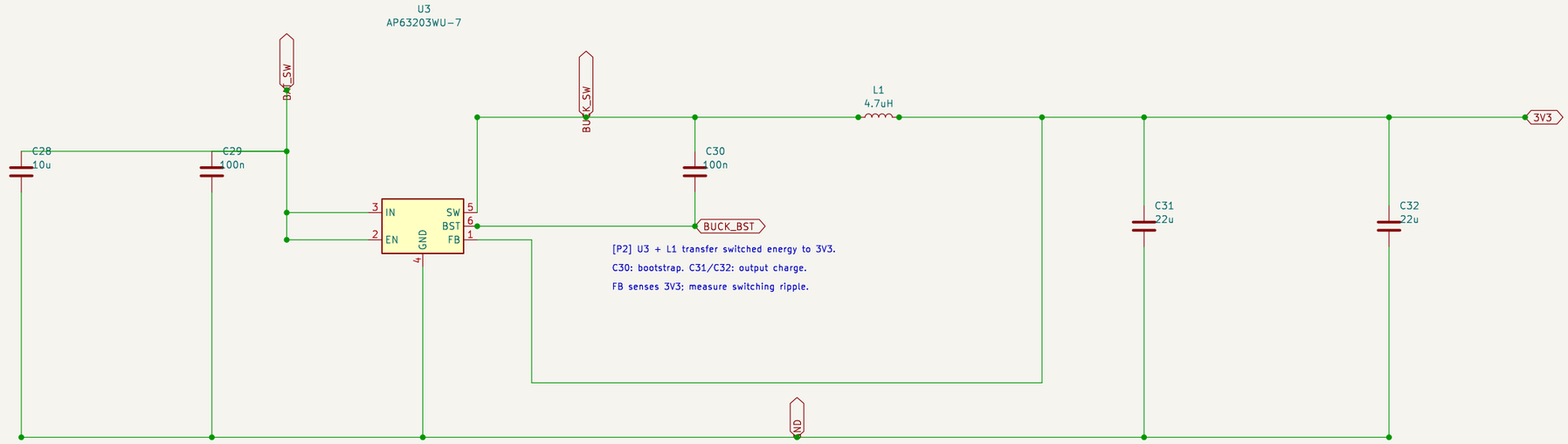
D6 blocks reverse polarity but loses forward voltage.

J4 disconnects BAT_SW; unplug the pack for master isolation.

Protected 2S pack / external charging only

J4 shunt enables power; remove shunt or unplug battery for storage

## 3.3 V / RADIO SUPPLY



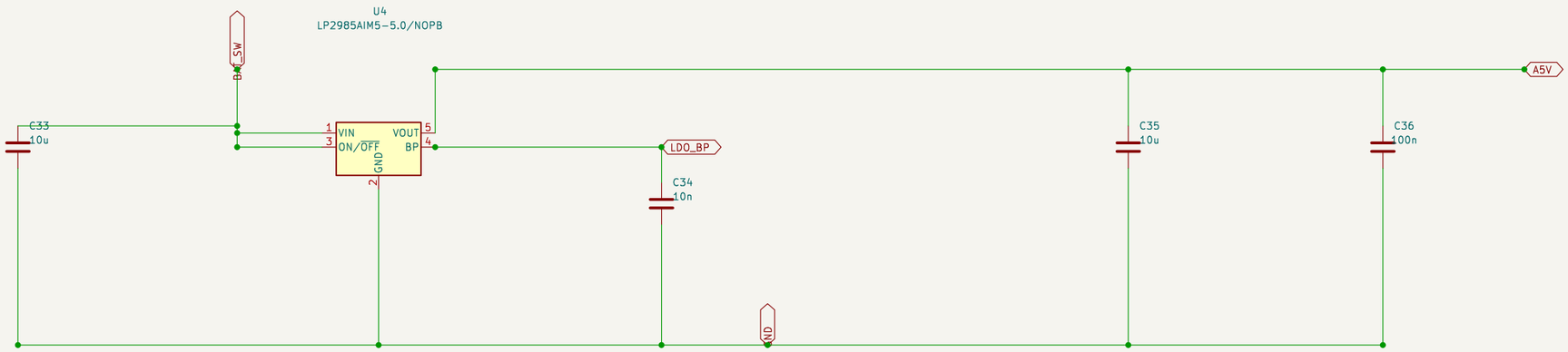

[P2] U3 + L1 transfer switched energy to 3V3.

C30: bootstrap. C31/C32: output charge.

FB senses 3V3; measure switching ripple.

[P3] Separate analog LDO avoids another local switch node; it is not galvanic isolation.

## 5 V / QUIET ANALOG SUPPLY

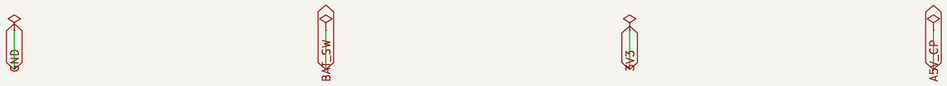



C34 bypasses the LDO noise-related node; C35 supports stability.

Check effective capacitance / ESR for the installed LP2985 revision.

Linear regulation trades switching activity for heat: P ~ (Vin-Vout) x I.

Power flags / ERC declarations




[P4] PWR_FLAG declares a driven rail for ERC; it adds no physical part.

The pack needs independent protection. No onboard charger or hardware low-voltage cutoff.

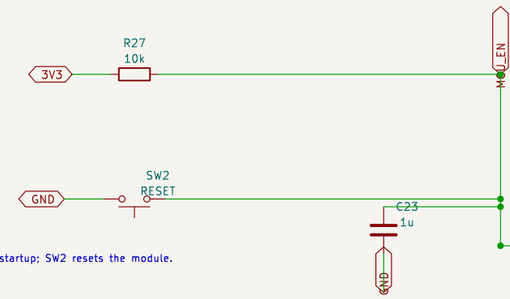
| | |
|---|---|
| Sheet: /Power/ | |
| File: Power.kicad_sch | |
| **Title: EEG learning prototype — Power** | |
| Size: A3 | Date: 2026-10-12 |
| KiCad E.D.A. 9.0.9.1 | **Rev: A — bench prototype** |
| | Id: 4/5 |

# ESP32-C3 / BLE DATA

[R1] U2 reads ADC frames on DRDY, buffers samples, then sends BLE packets.

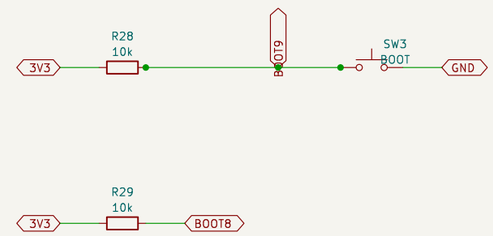
Acquisition timing and radio timing are separate tasks. Firmware is still pending.

Reset / enable RC




[R2] R27/C23 define EN startup; SW2 resets the module.

Boot / download straps




R28–R30 define boot straps; SW3 selects download mode.

Hold BOOT, tap RESET, release BOOT for UART download.

[R5] Divider: Vsense = Vbat / 4.3.

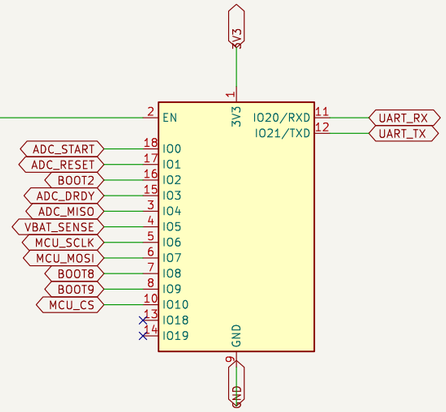
At 8.4 V: about 1.95 V and 19.5 uA.

C27 smooths / supports ADC sampling.

Calibrate and verify settling / attenuation.

Monitoring does not disconnect the pack.

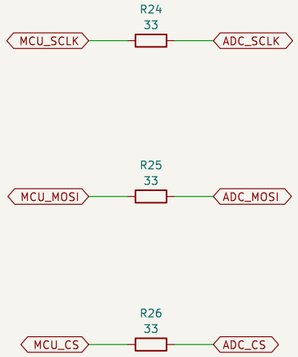
U2
ESP32-C3-WROOM-02-N4



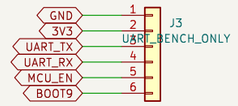

ADC SPI edge damping

[R3] R24–R26 damp fast SPI edges.

33 Ohm is signal damping, not electrode protection.




Bench UART

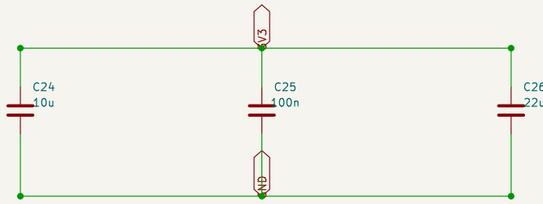



[R4] J3 is bench-only: 3.3 V logic UART.

Pin 2 senses the rail; do not feed adapter VCC.

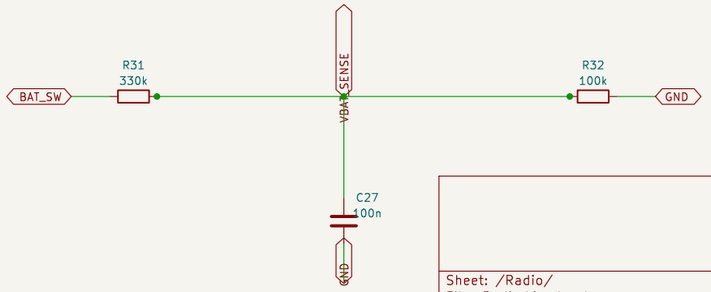
Disconnect electrodes before programming.

3.3 V local bypass




[R6] C24–C26 provide local charge for radio load changes.

Battery monitor / divider ratio 4.3




Sheet: /Radio/
File: Radio.kicad_sch

**Title: EEG learning prototype – Radio**

Size: A3 | Date: 2026-10-12 | **Rev: A – bench prototype**

KiCad E.D.A. 9.0.9.1 | Id: 5/5